

Photons, Atoms and Nuclei, solved

Step-by-step solutions to every practice problem in Chapter 24. Try each problem on your own before you read its solution.

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P24.1

Light is absorbed one photon at a time, and each photon gives all its energy to a single electron. Potassium has a work function of about 2.3 eV.

- A red photon (about 700 nm) carries $1240/700 = 1.8$ eV, which is less than 2.3 eV. It cannot free an electron, however many such photons arrive. More intense light means more photons, not more energetic ones, and an electron essentially never absorbs two photons at once.
- A blue photon (about 450 nm) carries 2.8 eV, which exceeds the work function. Each one can eject an electron, with up to 0.5 eV of kinetic energy to spare.

So the effect depends on frequency, and intensity only changes how many electrons are emitted.

P24.2

Baseball (0.145 kg at 40 m/s):

$$\lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34}}{0.145(40)} = 1.1 \times 10^{-34} \text{ m.}$$

That is about 10^{19} times smaller than a nucleus. There is no slit or obstacle small enough to diffract it, so its wave behaviour is unobservable, and classical mechanics works perfectly.

Electron with 1 eV of energy: $\lambda = h/\sqrt{2m_e K} = 1.2$ nm. That is comparable to atomic spacings, so crystals diffract electrons strongly.

P24.3

Ionisation requires an energy of roughly 10 eV or more, delivered to a single atom or molecule. Whether that can happen depends on the energy of *each photon*, $E = hf$, not on the total intensity.

- X-ray photons (keV) and gamma-ray photons (MeV) each carry thousands to millions of times the energy needed, so they ionise atoms and break bonds, including bonds in DNA.
- Radio and microwave photons carry only 10^{-7} to 10^{-3} eV each.

High intensity means many more of these weak photons. They can only jiggle molecules and so deposit heat. They cannot ionise. That is why radio-frequency safety limits are based on heating (SAR), while X-ray safety is based on ionisation dose.

P24.4

After two half-lives, $(\frac{1}{2})^2 = 25\%$ of the nuclei remain, so the sample is far from gone.

After ten half-lives, $(\frac{1}{2})^{10} = 1/1024 \approx 0.1\%$ remains.

Storage. Ten half-lives cuts the activity by about a factor of 1000. For short-lived medical isotopes such as Tc-99m ($t_{1/2} = 6$ h), that takes only 2–3 days. By then the residual activity is normally below the regulatory clearance level, and the waste can safely be disposed of as ordinary clinical waste. The method doesn't work for long-lived isotopes: ten half-lives of plutonium-239 is about 240 000 years.

P24.5

Energy per photon:

$$E = \frac{1240}{633} = 1.96 \text{ eV} = 3.14 \times 10^{-19} \text{ J}.$$

Photons per second:

$$N = \frac{P}{E} = \frac{1.0 \times 10^{-3}}{3.14 \times 10^{-19}} = 3.2 \times 10^{15} \text{ s}^{-1}.$$

Even a feeble laser emits quadrillions of photons a second, which is why light looks continuous in everyday life.

P24.6

Einstein's equation, $eV_s = hf - \phi$, is a straight line in f with slope h/e .

Frequencies: $f_{300} = c/\lambda = 1.00 \times 10^{15}$ Hz and $f_{400} = 7.50 \times 10^{14}$ Hz.

Planck's constant:

$$h = \frac{e \Delta V_s}{\Delta f} = \frac{(1.602 \times 10^{-19})(2.03 - 1.00)}{2.50 \times 10^{14}} = 6.6 \times 10^{-34} \text{ J s.}$$

Work function:

$$\phi = hf_{400} - eV_s = \frac{(6.60 \times 10^{-34})(7.50 \times 10^{14})}{1.602 \times 10^{-19}} \text{ eV} - 1.00 \text{ eV} = 3.09 - 1.00 = 2.1 \text{ eV.}$$

Robert Millikan made exactly this measurement in 1916, intending to *disprove* Einstein's theory, and confirmed it instead.

P24.7

$$\lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34}}{(1.675 \times 10^{-27})(2200)} = 1.8 \times 10^{-10} \text{ m} = 0.18 \text{ nm.}$$

Usefulness. This wavelength is comparable to the spacing between atoms, so neutrons diffract from crystals, just as X-rays do. But they complement X-rays:

- Neutrons scatter from *nuclei*, not from electrons, so they “see” light atoms such as hydrogen clearly. X-rays barely notice hydrogen, which matters a great deal for biomolecules and water.
- Neutrons can tell isotopes apart, such as hydrogen and deuterium.
- Neutrons have a magnetic moment, so they reveal magnetic structure.
- They penetrate deeply and do not damage samples by ionisation.

P24.8

Relativistic (as in Example 24.4):

$$pc = \sqrt{K^2 + 2Km_e c^2} = \sqrt{(200)^2 + 2(200)(511)} = 494 \text{ keV,} \quad \lambda = \frac{1240 \text{ eV nm}}{494\,000 \text{ eV}} = 2.51 \text{ pm.} \checkmark$$

Non-relativistic (wrong): $pc = \sqrt{2Km_e c^2} = \sqrt{2(200)(511)} = 452 \text{ keV}$, so

$$\lambda = \frac{1240}{452\,000} \text{ nm} = 2.74 \text{ pm}.$$

That is about 9% too long. Electron-microscope designers must use the relativistic formula.

P24.9

The Balmer lines end on $n_f = 2$, with $E = 13.6 \left(\frac{1}{4} - \frac{1}{n_i^2} \right) \text{ eV}$ and $\lambda = 1240/E \text{ nm}$:

TRANSITION	ENERGY	WAVELENGTH	COLOUR
$3 \rightarrow 2$ (H α)	1.89 eV	656 nm	red
$4 \rightarrow 2$ (H β)	2.55 eV	486 nm	blue-green
$5 \rightarrow 2$ (H γ)	2.86 eV	434 nm	violet

The higher Balmer lines crowd together towards the series limit at 365 nm, in the ultraviolet.

P24.10

(a) Half-value layer:

$$x_{1/2} = \frac{\ln 2}{60 \text{ cm}^{-1}} = 0.012 \text{ cm} = 0.12 \text{ mm}.$$

(b) Through 0.50 mm (0.050 cm):

$$T = e^{-60(0.050)} = e^{-3.0} = 5.0\%.$$

The apron blocks about 95% of 100 keV scatter. Most scattered radiation has lower energy, where lead attenuates even more strongly, so in practice the protection is better than this.

(c) For 0.10% transmission:

$$x = \frac{\ln 1000}{60} = \frac{6.9}{60} = 0.115 \text{ cm} = 1.2 \text{ mm}.$$

P24.11

(a) **Activity at 09:00.** The dose decays for 90 minutes before injection, so the 09:00 activity must be larger by $2^{90/110}$:

$$A_{09:00} = 370 \times 2^{90/110} = 370 \times 1.76 = 650 \text{ MBq.}$$

(b) **The annihilation photons.** The positron slows down to almost rest before annihilating with an electron, which is also nearly at rest. So the total momentum before annihilation is essentially zero. Momentum must be conserved, so the photons must carry equal and opposite momenta: they fly apart **back to back** at 180° . Energy conservation requires them to share the rest energy $2m_e c^2 = 1022 \text{ keV}$, so each carries 511 keV . Detecting both photons at once defines a line through the patient on which the annihilation occurred. Many such lines are combined to reconstruct the image.

P24.12

Expected activity for 0.50 g of living carbon: $0.23 \times 0.50 = 0.115 \text{ Bq}$. The measured activity is 0.050 Bq, so the ratio is

$$\frac{A}{A_0} = \frac{0.050}{0.115} = 0.435.$$

Age:

$$t = \frac{t_{1/2}}{\ln 2} \ln \frac{A_0}{A} = \frac{5730}{0.693} \ln 2.30 = 8267 \times 0.833 = 6.9 \times 10^3 \text{ years.}$$

P24.13

(a) **Maximum electron energy.** The electron gains the most when the photon bounces straight back ($\theta = 180^\circ$, so $1 - \cos \theta = 2$). A 511 keV photon has $\lambda = hc/E = 2.43 \text{ pm}$, which happens to equal the Compton wavelength $h/m_e c$. So

$$\lambda' = 2.43 + 2(2.43) = 7.28 \text{ pm}, \quad E' = \frac{511}{3} = 170 \text{ keV.}$$

The electron receives

$$K_{\max} = 511 - 170 = 341 \text{ keV.}$$

This is the *Compton edge*. A single Compton scatter can never deposit the full 511 keV.

(b) **Detector crystals.** To identify a genuine 511 keV annihilation photon, rejecting photons that scattered in the patient and lost energy, the detector should absorb the *whole* photon energy. That happens mainly through the photoelectric effect, which grows very rapidly with atomic number (roughly as Z^4 – Z^5). Dense, high- Z crystals such as LSO (lutetium oxyorthosilicate) and BGO (bismuth germanate) stop 511 keV photons within about 1–2 cm and absorb a large fraction of them fully. Plastic scintillators are made of low- Z carbon and hydrogen. They interact mainly by Compton scattering, so they record only part of the energy, and they are inefficient at stopping such energetic photons.

P24.14

(a) **Radius.** Coulomb's attraction provides the centripetal force:

$$\frac{ke^2}{r^2} = \frac{m_e v^2}{r}.$$

The quantisation condition gives $v = \frac{n\hbar}{m_e r}$. Substituting:

$$\frac{ke^2}{r} = m_e \left(\frac{n\hbar}{m_e r} \right)^2 \Rightarrow r_n = \frac{n^2 \hbar^2}{m_e k e^2} = n^2 a_0, \quad a_0 = \frac{4\pi\epsilon_0 \hbar^2}{m_e e^2}.$$

$$a_0 = \frac{(1.055 \times 10^{-34})^2}{(9.11 \times 10^{-31})(8.99 \times 10^9)(1.602 \times 10^{-19})^2} = 5.29 \times 10^{-11} \text{ m}.$$

(b) **Energy.** From the force equation, $\frac{1}{2}m_e v^2 = \frac{ke^2}{2r}$, so

$$E = \frac{1}{2}m_e v^2 - \frac{ke^2}{r} = -\frac{ke^2}{2r_n} = -\frac{ke^2}{2a_0 n^2}.$$

$$\frac{ke^2}{2a_0} = \frac{2.31 \times 10^{-28}}{2(5.29 \times 10^{-11})} = 2.18 \times 10^{-18} \text{ J} = 13.6 \text{ eV}.$$

So $E_n = -13.6 \text{ eV}/n^2$. ✓

P24.15

Mass before: $235.043930 + 1.008665 = 236.052595 \text{ u}.$

Mass after: $140.914411 + 91.926156 + 3(1.008665) = 235.866562 \text{ u}.$

Energy released:

$$\Delta m = 0.186033 \text{ u}, \quad Q = 0.186033 \times 931.5 = 173 \text{ MeV}.$$

(About 200 MeV in total once the later decays of the fragments are included.)

Per kilogram. One ^{235}U nucleus has a mass of $235 \times 1.66 \times 10^{-27} = 3.90 \times 10^{-25} \text{ kg}$, so

$$\frac{173 \times 1.602 \times 10^{-13} \text{ J}}{3.90 \times 10^{-25} \text{ kg}} = 7.1 \times 10^{13} \text{ J/kg}.$$

Comparison: $7.1 \times 10^{13} / 3 \times 10^7 \approx 2.4$ million times more energy per kilogram than coal. A 1 GW reactor uses about a tonne of uranium fuel a year, and only a few percent of that is actually fissioned. A coal plant of the same output burns about 3 million tonnes of coal.

P24.16

(a) Energy released.

$$\Delta m = (2.014102 + 3.016049) - (4.002603 + 1.008665) = 5.030151 - 5.011268 = 0.018883 \text{ u}.$$

$$Q = 0.018883 \times 931.5 = 17.6 \text{ MeV}.$$

Per kilogram of fuel, that is about 4 times more than fission.

(b) Coulomb barrier:

$$U = \frac{ke^2}{r} = \frac{(8.99 \times 10^9)(1.602 \times 10^{-19})^2}{3.0 \times 10^{-15}} = 7.7 \times 10^{-14} \text{ J} = 0.48 \text{ MeV}.$$

(c) Temperature for $kT = U$:

$$T = \frac{U}{k} = \frac{7.7 \times 10^{-14}}{1.38 \times 10^{-23}} = 5.6 \times 10^9 \text{ K}.$$

Why lower temperatures suffice.

1. The Maxwell distribution has a long high-energy tail, so a small fraction of nuclei have many times the average energy.
2. **Quantum tunnelling** lets nuclei penetrate the barrier without having enough energy to go over it, a direct consequence of their wave nature.

Together, these give useful fusion rates at about $1.5 \times 10^8 \text{ K}$ in reactors such as tokamaks, and about $1.5 \times 10^7 \text{ K}$ in the Sun's dense core, where the enormous number of nuclei makes up for the

very low rate per nucleus.

P24.17

(a) **Combined removal.** Radioactive decay removes the substance at rate $\lambda_p N$, and excretion removes it independently at rate $\lambda_b N$. So

$$\frac{dN}{dt} = -(\lambda_p + \lambda_b)N, \quad \lambda_{\text{eff}} = \lambda_p + \lambda_b.$$

Since $\lambda = \ln 2/T$, this gives

$$\frac{1}{T_{\text{eff}}} = \frac{1}{T_p} + \frac{1}{T_b}.$$

The effective half-life is always shorter than either one.

(b) **Iodine-131 in the thyroid:**

$$\frac{1}{T_{\text{eff}}} = \frac{1}{8.0} + \frac{1}{80} = 0.1375 \text{ d}^{-1} \Rightarrow T_{\text{eff}} = 7.3 \text{ d}.$$

Physical decay dominates, because the thyroid holds on to iodine.

Tc-99m tracer:

$$\frac{1}{T_{\text{eff}}} = \frac{1}{6.0} + \frac{1}{24} = \frac{5}{24} \text{ h}^{-1} \Rightarrow T_{\text{eff}} = 4.8 \text{ h}.$$

Encouraging patients to drink fluids and empty their bladder often shortens T_b , and with it the dose.

P24.18

(a) **Total decays.** The number of nuclei is $N = A/\lambda$, with $\lambda = \ln 2/(6.0 \times 3600 \text{ s}) = 3.21 \times 10^{-5} \text{ s}^{-1}$:

$$N = \frac{7.40 \times 10^8}{3.21 \times 10^{-5}} = 2.3 \times 10^{13} \text{ decays}.$$

(b) **Dose.** The energy absorbed is half of 140 keV per decay:

$$E = (2.3 \times 10^{13})(70 \times 10^3 \text{ eV})(1.602 \times 10^{-19} \text{ J/eV}) = 0.26 \text{ J}.$$

$$D = \frac{0.26 \text{ J}}{70 \text{ kg}} = 3.7 \times 10^{-3} \text{ Gy} = 3.7 \text{ mGy} \approx 3.7 \text{ mSv}.$$

The weighting factor for gamma rays is 1.

(c) **Comparison.** This is roughly 1.2–1.8 years' worth of natural background radiation (2–3 mSv per year). Biological excretion through the kidneys, ignored here, roughly halves the real figure. Published effective doses for a bone scan are about 3–5 mSv. Doctors weigh this small added risk against the diagnostic benefit, such as finding bone metastases or occult fractures, in line with ALARA.

All worked examples and practice problems in these notes are original. Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

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